

Shuhao LI

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INTRO

I am a first-year PhD student in Civil and Environmental Engineering at HKUST, and a recipient of the **Hong Kong PhD Fellowship Scheme (HKPFS)** and the **HKUST RedBird PhD Recruitment Award**. My research focuses on AI-driven optimization and control of building energy systems. I have published **10 SCI-indexed journal papers, including five as first author**, mainly in JCR Q1 journals such as *Energy and Buildings* and *Building Simulation*.

My research enables real-time cooling plant optimization and deployment across buildings. I use dynamic programming and other algorithms to accelerate optimization for real-time control, develop **AutoLFM** to automate building load forecasting model development, and use Brick semantic models to reuse control logic and reduce deployment effort. Field trials at a commercial complex in Shanghai achieved **11.52% energy savings after correction for outdoor wet-bulb temperature**. Alternating-week trials at HKUST from April to July 2026 achieved **7.57% system energy savings and a 5.12% improvement in system COP**.

I participate in the **SmartCool Optimizer project**, working on algorithm development, BMS/SCADA and API integration, operational monitoring and field validation. Our team received a **Gold Award** and the **Romanian Inventors Forum Special Prize** at Indonesia Inventors Day 2026.

EDUCATION

2026.09-Present	Hong Kong University of Science and Technology	<i>Ph.D. in Civil and Environmental Engineering</i>
	➤ Selected Honors: Hong Kong PhD Fellowship Scheme; HKUST RedBird PhD Recruitment Award	
2024.09-2026.08	Hong Kong University of Science and Technology	<i>M.Phil. in Civil Engineering</i>
	➤ MPhil studies; transferred to the PhD programme in September 2026.	
2020.09-2024.07	Southeast University	<i>Bachelor in Intelligent Building</i>
	➤ GPA 3.85/4.00	
	➤ Recipient of the <i>National Scholarship</i> (2022)	

AWARD

Gold Award & Romanian Inventors Forum Special Prize, IID 2026	Oct 2026
<i>SmartCool Optimizer team; Indonesia Inventors Day, Jakarta.</i> - Additional recognition: Royal Malaysian Air Force Certificate of Appreciation.	
Second Prize, 12th Hong Kong University Student Innovation and Entrepreneurship Competition	May 2026
<i>Entrepreneurship Category; Smart Cool Tech Limited.</i>	
Hong Kong PhD Fellowship Scheme	Apr 2026
Silver Award, Hong Kong Techathon+ 2026	Jan 2026
<i>Sustainability & ESG Track; Smart Cool Tech Limited.</i>	
HKUST RedBird PhD Recruitment Award	Jan 2026
National Scholarship (Top 0.2%)	Oct 2022
Excellent student cadre (Top 1%)	Sep 2022
Outstanding student title (Top 2%, three times)	Sep 2022
Jiangsu Province Construction Scholarship	Nov 2021

LEADERSHIP & ACTIVITIES

President, the Science and Technology Association of the School of Civil Engineering

Jun 2022 — Jun 2023

- Organized an inter-provincial competition with over 4,000 participants.
- Led a school-level organization with more than 100 members.

PATENT

1. Wang, Z., **Li, S.H.**, Li, S. Method for controlling a chiller and its control system, device, storage medium, and program product. CN121804052A; Chinese patent application 202511426158.X (published 2026).
2. Wang, Z., Li, S., **Li, S.H.** Chiller operation sequence optimization method, system, storage medium and product. CN118882286A; application 202410909230.3 (published 2024).
3. Wang, Z., **Li, S.H.**, Li, S. Chiller Sequencing Optimization Based on Data-Driven Model Predictive Control with Dynamic Programming Method. Hong Kong patent application 42026122281.4 (2026); same patent family as CN121804052A.

PUBLICATION

First Author

1. **Li, S.**, Li, S., Mohebi, P., Wang, D., Ma, M. N., Liu, G., & Wang, Z. (2025). Field demonstration of model predictive control for chiller sequencing in large-scale commercial buildings. *Energy and Buildings*, 344, 116021.
2. **Li, S.**, Su, S., & Lin, X. (2025). Optimizing the hyper-parameters of deep reinforcement learning for building control. *Building Simulation* (Vol. 18, No. 4, pp. 765-789). Tsinghua University Press.
3. **Li, S.**, Li, S., & Wang, Z. (2024). Accelerating chiller sequencing using dynamic programming. *Energy and Buildings*, 325, 115037.
4. **Li, S.**, Zheng, W., Li, M., Xing, S., & Wang, Z. (2026). AutoLFM: A multi-agent LLM framework for automated building load forecasting. *Building Simulation*, 19(2), 545–568.
5. **Li, S.**, Li, M., Hu, Z., & Wang, Z. (2026). Applying semantic model for easy and fast deployment of chiller sequencing algorithm. *Energy and Buildings*, 352, 116830.

Co-Author

6. Li, S., **Li, S.**, Liu, Y., Li, B., Zheng, C., & Wang, Z. (2026). Model predictive control of variable refrigerant flow systems for demand response: balancing the trade-off between model accuracy and optimization complexity: a case study of indoor system. *Energy Conversion and Management*, 357, 121400.
7. Mohebi, P., **Li, S.**, & Wang, Z. (2025). Chance-constrained stochastic framework for building thermal control under forecast uncertainties. *Energy and Buildings*, 331, 115385.
8. Su, S., **Li, S.**, Ding, Y., Mao, P., & Chong, D. (2024). Health damage assessment of commuters and staff in the metro system based on field monitoring—A case study of Nanjing. *Frontiers in Public Health*, 11, 1305829.
9. Zheng, W., Liu, L., **Li, S.**, Zhang, Z., Zhao, P., Li, T., ... & Wang, Z. (2026). Smart predict-then-optimize-based model predictive control for Virtual Power Plants with battery storage. *Journal of Energy Storage*, 142, 119606.
10. Li, M., Hu, Z., Mohebi, P., **Li, S.**, & Wang, Z. (2026). Enhancing LLM-based building data query with chain-of-thought, retrieval-augmented generation, and fine-tuning. *Automation in Construction*, 182, 106738.

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